

The Record, Not the Streak: An Ecosystem-Scale Study of Satisfaction in Islamic Prayer Applications

ANONYMOUS AUTHOR(S)

Mobile apps mediate daily prayer for many Muslims, yet research treats them mainly as habit trackers. We analyze 732,194 reviews from 26 Islamic prayer apps and find that framing misses what drives satisfaction. Streaks and completion scores, borrowed from fitness apps, do not induce guilt: guilt language is rarer in reviews discussing prayer tracking. Feature count does not predict complaints about clutter; what users call “bloat” is about what the interface shows on screen. And 41.6% of complaints about wrong prayer times or qibla direction come from four- and five-star reviews, so ratings hide serious defects. Satisfaction rests on reliable core functions, a trustworthy prayer record, and reminders that arrive, not on features or polish. Three lessons generalize: ratings are an unsafe proxy for defect severity, single-app sampling can manufacture false genre-wide patterns, and off-the-shelf NLP tools break on religious language.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *HCI design and evaluation methods*; Human computer interaction (HCI).

Additional Key Words and Phrases: app review mining, aspect-based sentiment analysis, religion and technology, Islamic prayer applications, devotional computing, self-tracking, gamification, feature bloat, mobile app quality

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1 Introduction

Salah, the ritual prayer performed five times daily, is obligatory for practising Muslims. Each of the five prayer windows moves with the sun and the observer’s location, and a few further windows depend on more computation — times when prayer is disallowed, or when a travelling or menstruating person is exempt. A large and growing population uses mobile apps to handle this: computing prayer times, sounding the call to prayer (adhan), pointing toward Mecca (qibla), and increasingly logging whether a prayer was performed. That makes these apps’ failures different in kind from a productivity app’s. A missed prompt is not a missed reminder; for a user relying on it, it can be a missed prayer.

The small existing literature on these apps, and the larger literature on the design patterns they borrow, offers three ways of thinking about what makes them succeed or fail. The self-tracking and gamification literature says streaks and completion scores — mechanics a prayer tracker borrows from fitness apps — create guilt and anxiety in users who cannot keep them up. Product design treats feature richness and interface minimalism as one trade-off, with a richer feature set winning and keeping users. And accuracy is often treated as competing with aesthetics, as if tolerance for a wrong prayer time and tolerance for a cluttered screen were the same preference. Each framing is plausible and has real evidence behind it in adjacent domains, and each turns out to mislocate where satisfaction with a devotional app actually comes from.

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What is missing is evidence at ecosystem scale. The closest existing work is small: a dissertation study of prayer-app design [4], a two-app heuristic comparison [2], and two single-application content analyses of review themes [14, 15]. None mines reviews across the ecosystem with aspect-level sentiment, and none tests the gamification-harm hypothesis in a devotional context.

We analyse 732,194 reviews across 26 Islamic prayer-time and devotional apps on the Google Play Store, addressing six pre-specified research questions: aspect-based sentiment across 27 aspects, topic modelling, demand-signal extraction, promise-versus-delivery comparison against store descriptions, and five statistical models under a shared false-discovery-rate correction, corroborated by two independently drawn qualitative coding passes.

Three of the six research questions overturn a framing from the literature. Gamifying a prayer tracker neither helps nor harms sentiment, and guilt language is *rarer*, not commoner, inside tracker discourse; what users resent is a record they cannot trust and a prompt that never arrives (RQ1, §4). Accuracy and interface complaints cost the same *as classes*, but that average hides a severity gradient, and a large share of accuracy complaints arrive inside four- and five-star reviews, invisible to a rating-based read of severity (RQ2, §5). And feature count does not predict interface-bloat complaints in either direction; bloat is a property of the interface, shown directly by bloat and interface complaints co-occurring far above base rate within individual reviews (RQ5, §8). Two further questions test whether distinctive features confer competitive advantage (they do not; users cite core reliability instead, RQ3, §6) and whether bio-spiritual needs such as menstrual accommodation are met (they are not, though with only three implementing applications we treat this as an unmet-need finding rather than a causal one, RQ4, §7). A sixth question synthesises these into two distinct failures — a coverage gap and a delivery gap that need different remedies (RQ6, §9).

This paper makes four contributions: an ecosystem-scale account of devotional-app quality across six pre-specified research questions, reporting every non-inferable term rather than dropping it; a reframing of gamification harm in religious practice as record integrity and prompting reliability rather than guilt; a demonstration that devotional register breaks off-the-shelf sentiment and aspect-extraction tooling; and two sampling hazards in app-store research, each demonstrated with numbers rather than asserted — star ratings are an unsafe proxy for defect severity, and single-application sampling can manufacture what look like genre-level themes.

2 Related Work

We draw on four strands: religious technology in HCI, gamification and self-tracking, app-store review mining as a method, and the literature on feature bloat.

2.1 Religious and Devotional Technology

HCI and CSCW have studied religious practice across several traditions. Abokhodair et al. [1] analyse 2.6 million tweets sharing Quran verses, combining topic analysis with a qualitative study of devotional goals. Wolf et al. [18] design for “uncontrollability” in a Christian blessing ritual, and Bhattacharjee et al. [5] interview Hindu migrants sustaining practice across displacement. None addresses prayer-time apps specifically or reviews at ecosystem scale.

Closer to our topic, Bellar [4] analyses 65 Catholic and Islamic prayer-app descriptions and finds reminder features central to design — which our RQ3 and RQ6 corroborate independently at a much larger scale. Azahari et al. [2] compare two named apps by expert evaluation, finding intrusive ads in the feature-rich one against simplicity in the minimal one. Our RQ5 revisits that pattern with corpus evidence and complicates it: we do not find that a fuller feature set predicts worse sentiment.

The closest methodological precedent is a pair of studies by Noor et al. [15] and Noor et al. [14], who apply content analysis to reviews of one Muslim-lifestyle app and find themes — prayer-time accuracy, adhan reliability, advertising — that recur throughout our own results. Both studies analyse a single app (279 and 2,459 reviews); ours analyses 26 apps and 732,194 reviews with aspect-based sentiment and mixed-effects models, which is what lets us tell an ecosystem-level finding apart from an artefact of one app’s release history (§4.4).

2.2 Gamification and Self-Tracking

A substantial literature documents that gamified self-tracking — streaks, points, badges — can produce real harm. Renfree et al. [16] show that streaks sustain new behaviour but create a dependency whose loss introduces fragility. In interview studies of wearable-tracker users, Humphreys et al. [12] and Demetrovics et al. [8] report anxiety, guilt, and loss of autonomy, and Bai et al. [3] find competition-based gamification increases stress. Against this, a meta-analysis by Six et al. [17] finds no difference in outcomes between gamified and non-gamified mental-health apps, so the harm is not universal.

RQ1 asks whether the same mechanism operates when the tracked behaviour is a religious obligation rather than a personal fitness goal, and finds that it largely does not (§4). We read this as a boundary condition rather than a refutation: the harm this literature documents may depend on the tracked behaviour being one the user set for themselves, which a prayer obligation is not.

2.3 App-Store Review Mining as Method

Mining app-store reviews is an established method. Dąbrowski et al. [9] survey 182 papers on the field, and Ciurumelea et al. [7] build a review taxonomy across 39 apps.

The closest precedent for our scale is Haggag et al. [11], who extract over five million reviews for 278 mHealth apps and classify them into 14 aspect categories. We follow their basic move — ecosystem-scale mining resolved to named aspects — but depart on one point that matters for RQ2: several of their findings are stated as shares of users awarding one star to an aspect, treating the star rating as a readable index of severity. Our corpus does not support that assumption (§5). Guluzade and Sas [10] pair a functionality audit of 27 eating-disorder apps with a content analysis of reviews; their two-source design, a matrix read against what reviewers say, is structurally our promise-versus-delivery comparison (§3.3), and the nearest analogue we have found to RQ4’s menstrual-exemption gap outside a religious setting.

Beyond the devotional domain, this paper contributes two risks that this literature has not, to our knowledge, quantified directly: star ratings systematically hide a class of substantive complaint (§5), and single-app qualitative sampling can manufacture themes that read as genre-level findings (§4.4).

2.4 Feature Bloat and Interface Complexity

That more features degrade user experience is widely assumed and rarely tested. Marzi [13] reviews the “over-featuring” literature and finds the phenomenon ranked among the top risks of new-product-development failure, alongside a lack of investigation of its causes. Blackler et al. [6] find over-featured products do produce negative emotional experiences, and that most users do not exercise most features. RQ5 tests the commonly assumed causal link — that a higher feature count predicts more bloat complaints — directly against corpus data, and finds it fails in both directions (§8).

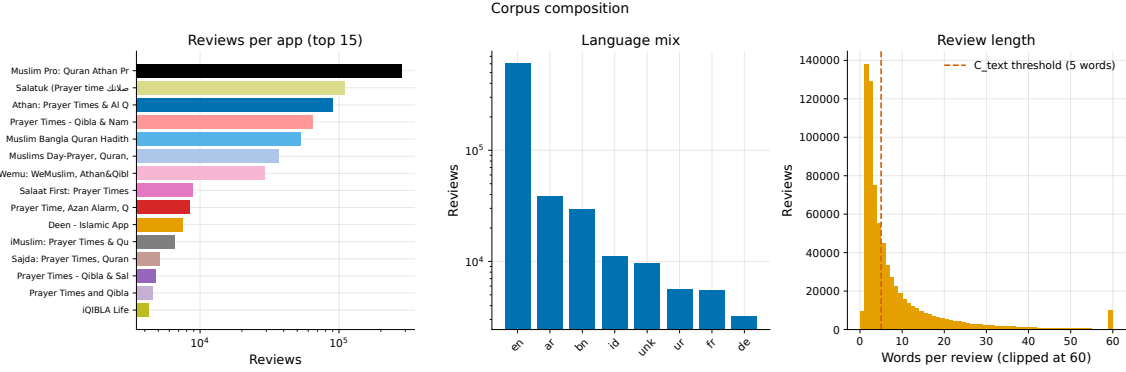


Fig. 1. Corpus composition: review volume for the 15 largest applications, language mix, and review-length distribution.

3 Method

3.1 Data Collection

We collected reviews for 26 Islamic prayer-time and devotional apps from Google Play on 2026-08-19, covering each app’s review history to that date, along with each app’s store description, install count, and rating. This gave 732,194 reviews (Figure 1).

We chose apps purposively: we prioritised high install and review counts for app-level comparison, favoured apps whose main purpose is salah (prayer times, adhan, qibla, tracking) over general Islamic-lifestyle apps, and kept a few smaller apps that were the only example of a feature category. This is not a census or random sample; our claims are about these 26 apps, which we take to represent the high-traffic core of the category.

Reviews were pulled with the store locale set to English and the United States, newest first. This does not mean the corpus is entirely in English — 48,265 of 324,687 texted reviews turned out to be non-English — but it does mean we see each app’s English/US-locale slice, not its global review population. Every app was collected the same way, so comparisons between apps remain fair; what the decision limits is generalisation (§11).

Filtering to reviews with text left 324,687 (C_{text}). Running fastText language identification on these gave an English subset of 276,422 reviews (C_{en}), used for topic modelling.

3.2 Sentiment and Aspect-Level Analysis

We scored every review’s sentiment two ways: a lexicon method (VADER) and a transformer classifier (RoBERTa), reported throughout. Across all 324,687 texted reviews (Figure 2), RoBERTa agrees with the star rating at Cohen’s $\kappa = 0.466$, more than twice VADER’s 0.210. On a 500-review gold set hand-labelled into four categories, two annotators agreed on 89.0% of 100 double-coded rows (Krippendorff’s $\alpha = 0.824$) — a measure of labelling consistency, not classifier accuracy.

To track specific complaints and praise, we built a lexicon of 27 aspects covering functional and experiential features (prayer-time accuracy, adhan reminders, Quran audio, qibla, tracking, ads, monetisation, and more), using 340 keywords and 106 regex patterns applied sentence by sentence. We release the full lexicon so readers can see exactly what triggers each tag.

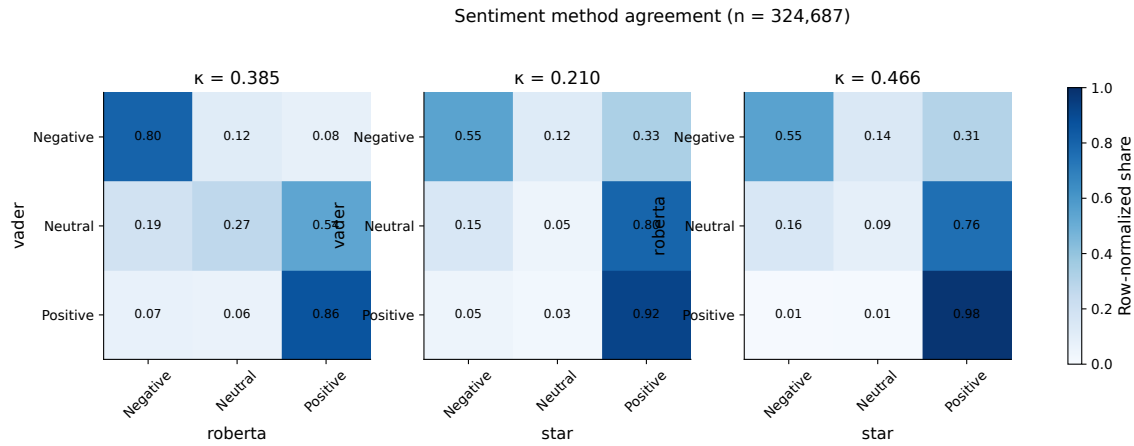


Fig. 2. Pairwise agreement between the three sentiment signals across the 324,687 texted reviews, row-normalised, with Cohen’s κ per pair. RoBERTa tracks the star-derived label more closely ($\kappa = 0.466$) than VADER does ($\kappa = 0.210$).

Aspect keywords built for ordinary product reviews misfire on devotional language. The clearest case: `goal_system` scored 0% precision because reviewers often close a review with “May Allah reward you,” which its keywords mistake for feedback about a goal-tracking feature. We removed aspect tags whose triggering sentence was this kind of devotional formula — 1,686 of 192,874 tagged (aspect, sentence) pairs (0.9%), leaving 191,188. We treat this as a real finding, not a cleanup step: general-purpose sentiment tools do not transfer cleanly to religious text.

For RQ1 we also built two small keyword lists, one for guilt/anxiety language and one for motivation language. These have no gold-standard validation, so we treat what they produce as a lexical-frequency comparison, not a validated measure of guilt, and rest RQ1’s main claim on hand-coded review passes instead (§4).

3.3 Demand Signals and Promise Extraction

We used pattern matching to flag five kinds of demand signal, each tied to the aspect it discusses: explicit requests, statements that a feature is missing, churn (“I’m leaving because...”), stated preference (“that’s why I use this app”), and long-term use. Against a 200-row gold sample, precision was 91.5% overall. This gold sample was labelled against an earlier extractor run; a later threshold change means only 11 of the 200 rows still match today’s extraction, so 91.5% is our best estimate rather than an exact figure for the current data.

To compare what apps promise against what they deliver, we extracted claimed features from each store description and checked them against a hand-built matrix (one row per app, one column per feature). Agreement is weak — mean Cohen’s $\kappa = 0.272$ across 20 features, with four at $\kappa = 0$ purely because one source records nearly every app as having, or lacking, that feature, leaving no variance for κ to work with (raw agreement on those same features still runs above 96%). Mean raw agreement across all 20 features is 73.1%. Figure 3 gives the full breakdown.

3.4 Topic Modelling

We fit a BERTopic model over the 276,422-review English subset, using a seeded UMAP reduction and a stopword list extended with Islamic terms and app names. This produced 40 topics plus an outlier bucket: one large undifferentiated “generic praise” topic (168,054 documents, 61%), an outlier bucket (62,745, 22.7%), and 39 labelled topics covering the

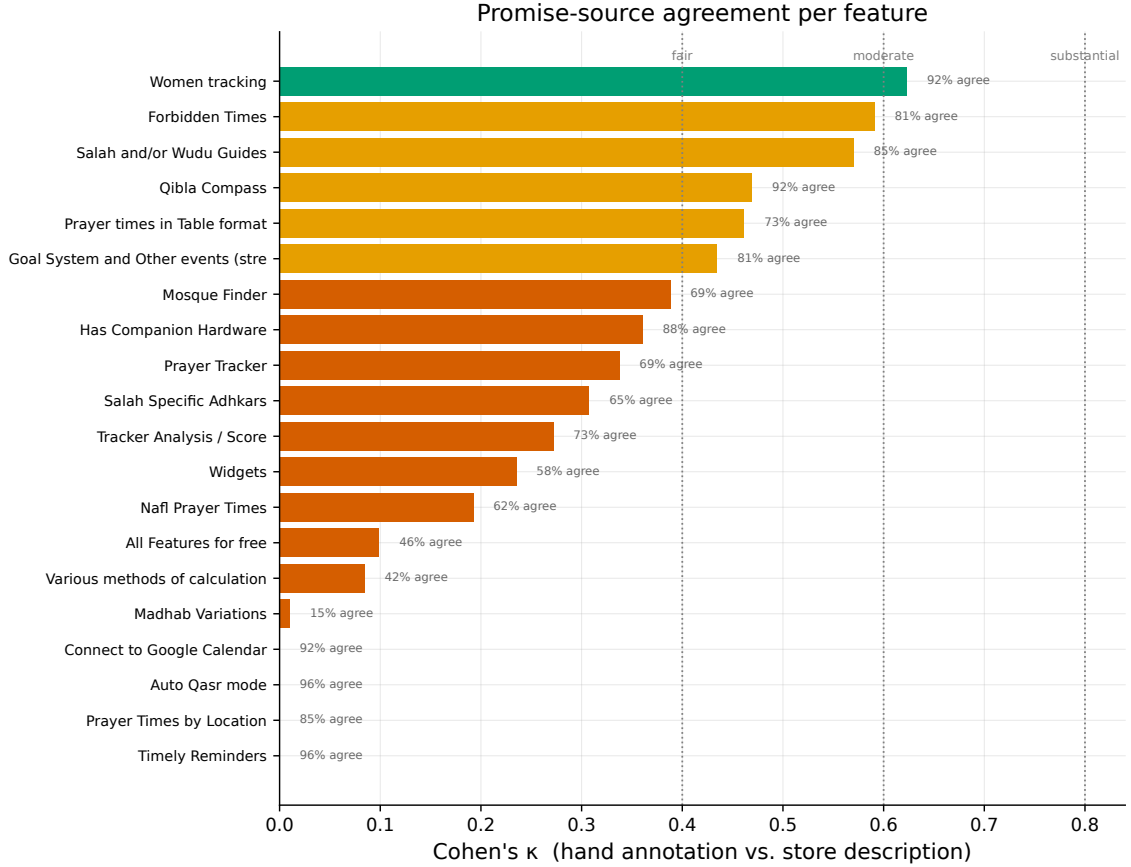


Fig. 3. Agreement between store-description promises and the hand-built feature matrix, by feature.

remaining 16%. All topics were hand-labelled. We treat the topic model as describing only the interpretable minority of the corpus.

We fit three targeted sub-models for tracker discourse (RQ1), menstrual and inclusion discourse (RQ4), and bloat discourse (RQ5). Because BERTopic's output is sensitive to random seed, we refit each across five fixed seeds (42–46) before inspecting any result. The tracker sub-model is stable (mean pairwise Adjusted Rand Index 0.837); the bloat sub-model is not (6–11 topics across seeds, ARI 0.452), so we treat its output as recurring themes rather than a fixed partition.

3.5 Statistical Models

We fit five mixed-effects regression models (M1–M5), each with a random intercept per app to account for reviews clustering within 26 very different apps: M1 (RQ2, cost of accuracy versus interface complaints), M2 (RQ1, sentiment by tracker tier), M3 (RQ4, inclusion), M4 (RQ5, bloat and hardware), M5 (RQ3, competitive-edge features). RQ6 is answered descriptively and enters no inferential model. Where a mixed model failed to converge, we report a cluster-robust OLS fit on the same data as a sensitivity check.

Table 1. Aspect-tagging precision against gold-standard rows, by aspect, for the 25 aspects whose gold rows survive the current lexicon configuration. Bold rows fall below 50% precision; any count for these is reported elsewhere only as a precision-adjusted estimate with its interval. Gold samples are small, so the intervals are wide and we report them rather than the point estimate alone.

Aspect	Corpus n	Gold n	Precision	95% CI	Adjusted estimate
reminders_adhan	30,069	7	100%	[65, 100]	
quran_audio	24,964	6	100%	[61, 100]	
prayer_times_accuracy	21,941	11	73%	[43, 90]	
ads_intrusive	17,845	11	100%	[74, 100]	
ui_design	15,720	11	73%	[43, 90]	
monetization	13,914	11	82%	[52, 95]	
adhkar	9,280	3	100%	[44, 100]	
qibla	8,526	11	82%	[52, 95]	
stability_bugs	7,665	11	64%	[35, 85]	
privacy_data	7,471	10	90%	[60, 98]	
prayer_tracker	6,009	9	78%	[45, 94]	
complexity_bloat	5,389	9	67%	[35, 88]	
widgets	4,755	11	100%	[74, 100]	
spiritual_affect	3,395	2	100%	[34, 100]	
mosque_finder	3,043	11	27%	[10, 57]	≈ 830 [297, 1,721]
companion_hardware	2,512	5	40%	[12, 77]	$\approx 1,005$ [295, 1,932]
women_period	2,194	10	20%	[6, 51]	≈ 439 [124, 1,119]
forbidden_times	1,534	10	10%	[2, 40]	≈ 153 [27, 620]
table_format	1,483	9	44%	[19, 73]	≈ 659 [280, 1,088]
nafl_times	963	5	100%	[57, 100]	
calc_method	639	10	90%	[60, 98]	
qasr_travel	637	2	0%	[0, 66]	do not report volume
madhab	491	5	60%	[23, 88]	
guides	202	1	100%	[21, 100]	
tracker_score	101	1	0%	[0, 79]	1 gold row, uninformative
Total	190,742	192			

Coefficients from the five primary models enter one Benjamini–Hochberg false-discovery-rate correction: 29 tests total, 20 surviving at $q < 0.05$. Eight terms in that set are technically non-converged — huge standard errors, $p \approx 0.99999$ — but we keep them in the denominator for transparency; they account for eight of the nine terms that fail correction. A further three return no standard error at all and sit outside the correction entirely. The cause is structural: a random intercept per app, fit over only 26 apps, cannot be separated from an app-level predictor in the same model. We cite a null result only from a model that actually converged.

3.6 Measurement Quality

Because several results depend on aspect tags, we checked their precision against a 300-row gold sample (11 rows per aspect). Of these, 192 still match under the current lexicon; unweighted precision across aspects is 69.3%, and corpus-volume-weighted precision is 84.4% (95% CI [79.7, 88.9]). Table 1 breaks this down by aspect. Seven aspects fall below 50% precision, two at 0%; every count drawn from one of these seven is reported elsewhere only as a precision-adjusted estimate with a confidence interval, never as a raw tagger count. Seventeen aspects sit at 64% or higher and supply almost all model input, with two flagged exceptions: madhab (60%, used in M1, and the one M1 term that reverses sign under a robustness check) and the mosque-finder tagger used in RQ3 (27%, where we read only the direction of the effect, not its size).

3.7 Ethics, AI Use, and Reproducibility

The corpus is public store review text; no users were recruited or contacted, so this study needed no human-subjects review. We still treat it as data about people: some reviews disclose menstruation, illness, or lapses in practice. We report no reviewer names, quote verbatim rather than paraphrase, and choose quotations with no incidental personal detail. Two quotations disclose menstruation, chosen because RQ4’s claim rests on users voicing that need explicitly. We name applications throughout, including in criticism, since they are public commercial products and the argument requires saying which app did what.

All analysis code was written with the assistance of Claude Code, an AI coding agent, used interactively by the authors. No generative model touched the review text, labels, or reported statistics; every number here comes from code run over the scraped corpus. The six research questions were pre-specified in a written analysis plan before analysis began, and all gold-standard labels were assigned by hand. One coding defect was found and fixed during analysis and did not affect any reported statistic.

The analysis was run on 2026-09-03 at commit 5da2db7; a from-scratch re-run reproduced byte-identical output. We release the full pipeline, aspect lexicon, feature matrix, gold and topic sheets, model output, and run manifest at a URL supplied on acceptance, withheld here for anonymous review. We do not release the raw review corpus, per the store’s redistribution terms.

4 RQ1: Gamification

The self-tracking literature says gamified mechanics — streaks, scores, badges — create guilt and anxiety in users who fall behind. Prayer trackers add exactly these mechanics to the five daily prayers, so RQ1 asks whether this happens here. It does not, at least not in the form the literature predicts. Gamifying a prayer tracker does not move sentiment either way, and guilt language is actually rarer in tracker discourse than elsewhere in the corpus. What users resent instead is a record they cannot trust and a prompt that never arrives.

4.1 Scope

We define “tracker discourse” as any review carrying one of three aspect tags: `prayer_tracker`, `tracker_score`, or `goal_system` — 6,285 reviews across 25 of the 26 apps. `prayer_tracker` (78% precision) accounts for 5,907 of these; the other two tags, both far noisier (0% precision against their gold rows), contribute the remaining 378 (6.0%). Noise at this level weakens a distributional signal rather than creating one. The coding passes below draw their own separate samples and do not depend on this boundary.

4.2 No Effect from the Tracker Mechanic, but a Bimodal Split

Comparing sentiment across three tiers — no tracker, a plain tracker, a tracker with a numeric score — the primary mixed-effects model failed to converge, so we report a cluster-robust sensitivity fit instead. Neither tier moves sentiment meaningfully: a plain tracker gives $\beta = -0.087$ (95% CI $[-0.332, 0.158]$) and a scored tracker $\beta = -0.134$ ($[-0.415, 0.147]$), both intervals comfortably spanning zero.

But a mean was never the right test. The literature’s love/hate prediction is about the *shape* of the sentiment distribution, not its centre, and Hartigan’s dip test confirms exactly that shape: $D = 0.118$, $p < 10^{-16}$ (Figure 4). Tracker sentiment splits toward the two poles, users loving the tracker or resenting it, with little middle ground — which is why the qualitative coding below, not the tier comparison, carries the real answer to RQ1.

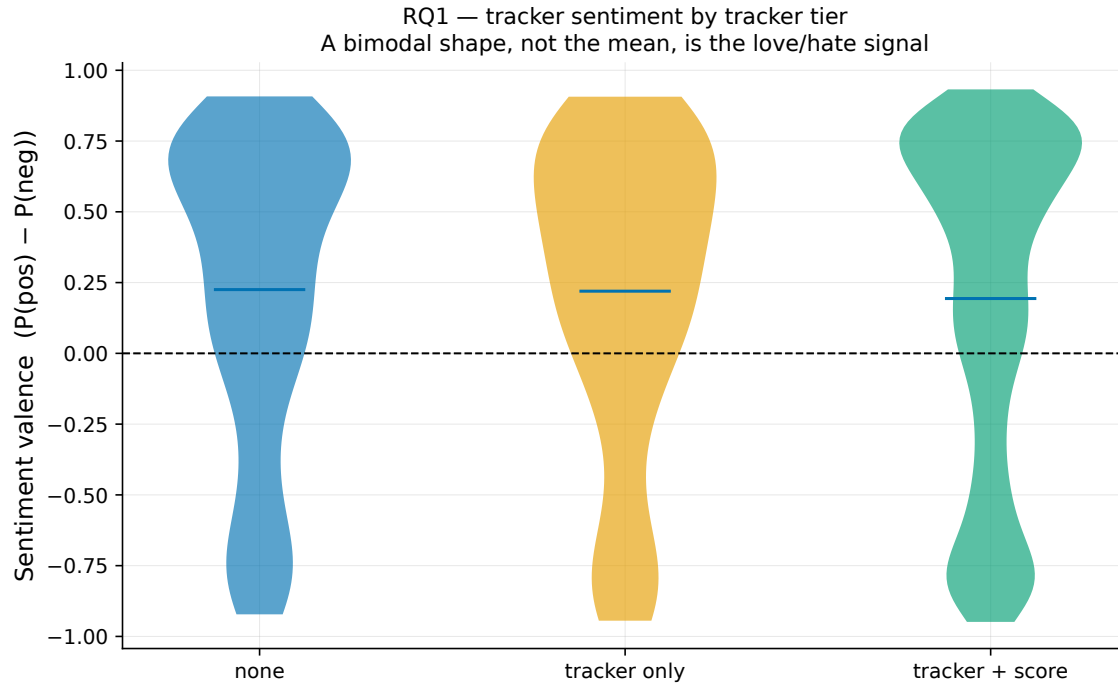


Fig. 4. Distribution of tracker-aspect sentiment valence, split by application tracker tier. Hartigan’s dip on the pooled tracker set gives $D = 0.118$, $p < 10^{-16}$.

4.3 Guilt Is Rare, Not Common

If streak anxiety were driving the negative half of that split, guilt language should appear more often in tracker discourse than elsewhere. It does not. Matching guilt/anxiety keywords against motivation keywords, the ratio inside tracker reviews is 0.26, against 0.86 elsewhere in the corpus (Fisher exact OR = 0.283, $p = 3.5 \times 10^{-11}$) (Figure 5). These lists were never gold-validated, so we treat this as a fact about word frequency, not a validated guilt measurement — the real evidence is the hand coding below, which points the same direction. Where guilt-adjacent language does appear, it is often light-hearted, folded into praise:

“I love the app. My everyday friend who reminds me to pray, and makes sure that I feel guilty if I don’t [sweat-smile emoji] I recommend to everyone who wants to start praying, or keep the habit in a consistent manner!”

— Athan: Prayer Times & Al Quran, 5 stars

4.4 Two Independent Coding Passes

We hand-coded two samples of reviews expressing tracker dissatisfaction. The first pass (50 reviews) came entirely from one app, Athan, an artefact of a selection-rule tie-break rather than evidence that one app dominates complaints. The second pass (30 reviews) corrected this, sampling across the other 22 apps. Pass 1 yielded 33 open codes ($n = 49$

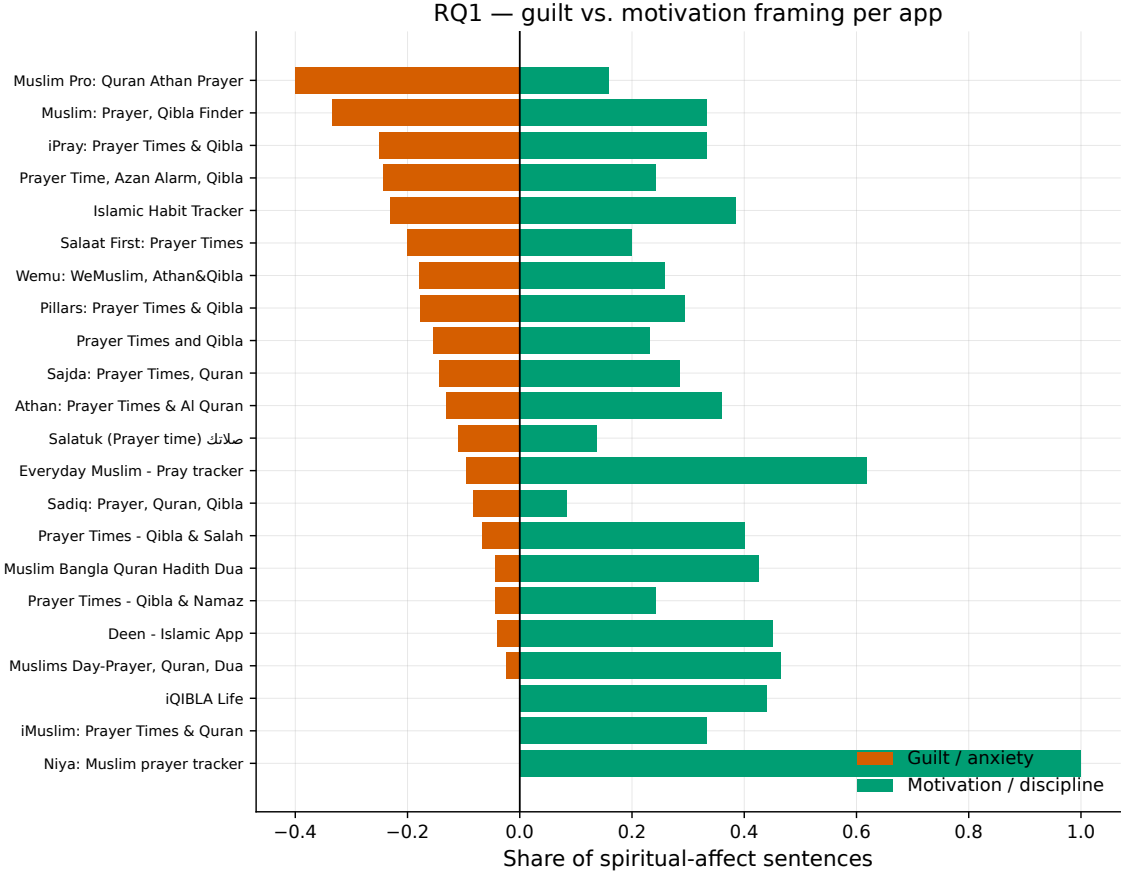


Fig. 5. Guilt- and motivation-coded affective language per application, as shares of that application's affect-coded reviews.

Table 2. Theme prevalence, coded independently in each pass. Never pooled.

Theme	Pass 1 (Athan, $n=49$)	Pass 2 (22 apps, $n=28$)
C. Record can't be trusted	11 (22%)	6 (21%)
D. Normative mismatch	6 (12%)	3 (11%)
E. Prompting failures	7 (14%)	13 (46%)
A. Redesign as regression	10 (20%)	0
B. Retroactive logging denied	5 (10%)	0
F. Monetisation vs. religious purpose	10 (20%)	1 (4%)
G. Record gated, not owned	—	3 (11%)*
H. App overreaches into device	—	2 (7%)*

*new theme, not present in the pass-1 frame

after exclusions); pass 2 yielded 25 codes ($n = 28$) plus two new themes. We never pool the two passes, since that would silently re-weight the ecosystem toward Athan (Table 2).

Guilt or judgment itself is rare in both passes: 2 of 49 coded complaints in pass 1 (4.1%), 1 of 28 in pass 2 (3.6%). A hypothesis this prominent in the self-tracking literature showing up at 4% or below, in two independently drawn samples, looks like a real null result, not an underpowered one.

The two passes also show a broader hazard for review-mining studies: themes A, B, and F (redesign, retroactive logging, monetisation versus religious purpose) together make up 51% of the single-app pass’s complaints but only 4% of the cross-app pass’s — single-app sampling manufacturing what looks like a genre-wide theme.

4.5 What Dissatisfaction Is Actually About

Three themes replicate across both samples, in the same rank order.

Prompting failure dominates, at 46% of the cross-app pass. The tracker does not fail because logging is tedious — it fails because the app never reliably prompts: location detection breaks, adhan playback does not fire, notifications get lost, calendars drift out of sync.

“the adhan is not working unless I open the app without logging out of it till prayer time”

— Pray Watch, 1 star

Record trust replicates almost exactly, at 22% in pass 1 and 21% in pass 2: users lose logged history, watch counts regress, or the tracker silently disables itself — often inside reviews that otherwise praise the app.

“I would give it a 5 star, but the prayers that I log on the previous days or current day would sometimes disappear. Please fix this!”

— Athan: Prayer Times & Al Quran, 4 stars

Normative mismatch is the smallest replicating theme, at roughly 11% in both passes, and it is not guilt. Where a review raises a genuine normative complaint, it is that the app *models worship incorrectly* — no menstrual exemption, no supererogatory (*sunnah*) prayer support, imposed goals that override the user’s own targets:

“It’s a really good app but I’m taking 1 star away cause I wish there was some option for menstruating women cuz my overall score on prayer tracker becomes really low when I can’t pray for my period.”

— Deen - Islamic App, 3 stars

Menstrual-exemption complaints turn up independently in both passes, from different apps — direct evidence we return to in RQ4 (§7).

4.6 One Application’s Episode

Two of the six original themes, redesign-as-regression (20% of pass 1) and retroactive-logging-denied (10%), disappear entirely in the cross-app pass. Both trace to a single Athan redesign that broke an existing logging workflow and removed retroactive logging. We report this as an application-specific episode, not an ecosystem theme — folding it in would overstate how much tracker dissatisfaction is really about record integrity in general. Overall, tracker dissatisfaction is about the app failing to prompt reliably and failing to keep a trustworthy record, and where a normative objection surfaces, it concerns the app’s model of worship rather than pressure to perform.

Table 3. M1 coefficients on star rating, all complaint aspects surviving BH-FDR at $q < 0.05$ ($n = 324,687$).

Complaint	β	95% CI
Intrusive ads	-1.439	[-1.460, -1.418]
Prayer-time accuracy	-1.112	[-1.142, -1.083]
Stability / bugs	-0.938	[-0.965, -0.910]
Reminders / adhan	-0.845	[-0.868, -0.823]
UI design	-0.600	[-0.664, -0.537]
Calculation method	-0.600	[-0.769, -0.430]
Qibla	-0.550	[-0.594, -0.507]
Madhab	-0.324	[-0.526, -0.122]

5 RQ2: Accuracy Versus Interface

Review-mining literature often treats accuracy and interface polish as one trade-off users weigh against each other. RQ2 asks whether users punish inaccuracy more harshly than a poor interface. As a class, they do not — but that is the wrong comparison: the average hides a steep severity gradient inside “accuracy,” and star ratings conceal the worst of it.

5.1 As Classes, They Cost About the Same

Using M1 ($n = 324,687$; all coefficients below survive Benjamini–Hochberg correction at $q < 0.05$), the accuracy family — prayer-time accuracy, qibla, madhab (the school of Islamic law whose rules set the prayer times), and calculation method — averages -0.647 stars against -0.600 for interface complaints, a gap under a twentieth of a star.

5.2 But Accuracy Is Not One Thing

Table 3 and Figure 6 show why. Prayer-time accuracy alone costs -1.112 stars (95% CI $[-1.142, -1.083]$), 1.85 times the interface penalty and one of the costliest complaints in the whole corpus. The family average is dragged down by milder, rarer complaints: madhab (-0.324) and qibla (-0.550). Getting the prayer time wrong is among the most costly failures we measure; getting the madhab setting wrong is among the least, and its coefficient even reverses sign under an ordinal robustness check ($\beta = +0.216$, $p = 0.649$), so we place no weight on it either way.

“this app is not accurate, prayers time are 1 hour ahead, please fix it”

— Athan: Prayer Times & Al Quran, 1 star

Qibla complaints, on inspection, are mostly about the compass, not the calculation: the larger of our two qibla topics (1,771 reviews) describes the compass giving a different bearing as the phone rotates in place — a sensor and calibration problem, not a math error.

5.3 Star Ratings Hide the Worst Accuracy Defects

This is RQ2’s strongest result, and it is methodological. Of 7,942 reviews with a negative accuracy complaint, 3,306 (41.6%) gave four or five stars, and 1,830 (23.0%) gave the maximum of five. Users routinely report a wrong prayer time or qibla direction and still rate the app highly in the same review. Any process that reads defect severity off the star distribution — which is how most app-store ranking and triage works — will systematically under-weight exactly this class of defect.

One more coefficient is worth flagging: intrusive advertising costs -1.439 stars, more than any accuracy complaint. We return to advertising in RQ5 and RQ6.

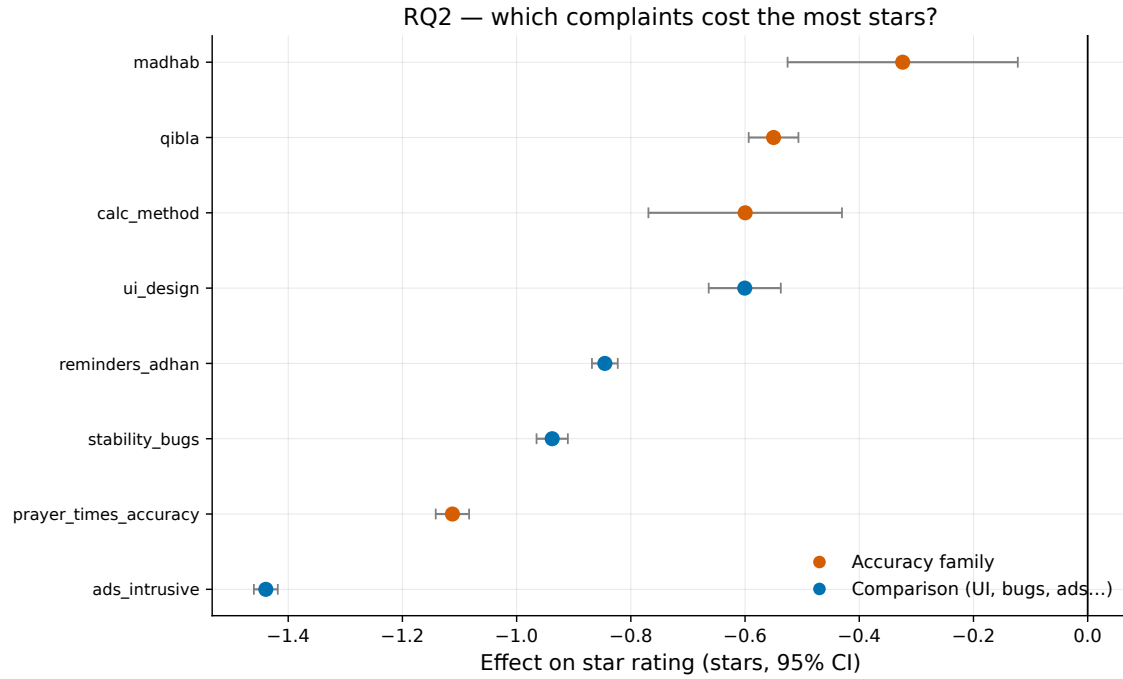


Fig. 6. M1 coefficients by complaint aspect, colour-coded by family. The spread within the accuracy family is visible directly in the error bars.

6 RQ3: Travel and Competitive Edge

A common claim in mobile-app design is that distinctive features — a mosque finder, a travel prayer-shortening (*qasr*) mode — win and keep users. RQ3 tests this against what users actually cite as reasons for loyalty. Neither feature supports the differentiation story: what earns loyalty here is reliable core function, not travel support.

6.1 What Users Actually Cite

Across 4,696 explicit preference citations (“that’s why I use this app”), the top aspects are adhan reminders (374), Quran audio (293), prayer-time accuracy (290), qibla (155), and adhkar, or remembrance prayers (152). Mosque finder does not make the top eight; *qasr* does not appear at all. Preference rates are nearly identical whether an app ships a mosque finder (1.38%) or not (1.24%).

6.2 Mosque Finder’s Coefficient Runs Backward, and Why

A cluster-robust fit on reviews mentioning mosque finder gives $\beta = -0.137$ ($p = 0.003$, $n = 3,010$): apps that *have* the feature draw more negative mentions of it, not fewer (27.0% versus 23.0%). The tagger is noisy (27% precision, 95% CI [10, 57]), so we read only the direction of the effect, not its size. The direction has a simple explanation: apps without the feature mostly get *requests* (“please add nearby mosques,” 243 signals, mostly neutral-to-positive), while apps that ship it get *usage reports* of a feature that sometimes fails, which read negative. A user can only complain about a feature that exists. The gap measures wishing for a feature versus using one, not the cost of shipping it. The coefficient is also

unstable across specifications — the mixed-effects fit on the same contrast returns an undefined standard error — so we rest no directional claim about competitive advantage on it.

6.3 Qasr: Demand Without a Test

Only one app of 26, *Muslim Bangla Quran Hadith Dua*, implements qasr, so no between-app comparison is possible — this is demand evidence, not a causal test. Demand is real but modest: 21 signals total (10 requests, 5 preference citations, 3 statements of absence, 3 churn statements), mostly from apps that lack the feature:

“My second suggestion is that prayer times automatically adjust when I’m traveling because man forgets and my prayers be off.”

— Muslim Pro: Quran Athan Prayer, 4 stars

A broader aspect tag for qasr matches 637 mentions but scored 0 of 2 on its gold rows and is evidently catching general travel and pilgrimage discourse rather than qasr specifically, so we do not report it as a measure of qasr volume and rest the argument on the 21 hand-read demand signals.

Where travel failures do appear, the pattern is prayer times not updating when location changes:

“...it doesn’t seem to automatically change your location when traveling until you physically refresh the prayer times page. Unfortunately today I accidentally broke my fast 1 minute early b/c the location was still set to an area almost 1 hour away, pls fix this inshaAllah.”

— Athan: Prayer Times & Al Quran, 4 stars

Neither travel-oriented feature confers a competitive edge. What users do cite — adhan reminders, Quran audio, prayer-time accuracy — points back to core functions, and those are themselves among RQ6’s most-broken promises (§9). Competitive advantage here looks like doing the basics reliably.

7 RQ4: Bio-Spiritual Inclusion

Islamic law exempts menstruating women from prayer; a tracker that does not represent this exemption misrepresents the practice it claims to track. RQ4 asks how well this ecosystem serves that need. Almost no one has built it, the few implementations are not trusted, and demand is voiced explicitly across nearly the whole ecosystem. This is an unmet need, not a contested design choice.

7.1 Almost No One Has Built It

The `women_period` tagger flags 2,194 sentences across 2,182 reviews in 24 of 26 apps (Figure 7), but its precision is only 20% (95% CI [6, 51]), so the adjusted estimate is closer to 439 genuine mentions. Only three apps — Athan, Islamic Habit Tracker, and Pillars — ship any menstrual handling at all.

7.2 Demand Is Explicit

Demand-pattern extraction, validated at 91.5% precision (§3.3), is firmer ground. The 272 demand signals tagged to `women_period` break down into 161 requests, 51 preference citations, 36 churn statements, 21 statements of absence, and 3 tenure citations, mostly from apps lacking the feature:

“a feature for sisters who are on their monthlies would be nice so they can keep track”

— Muslim Pro: Quran Athan Prayer, 4 stars

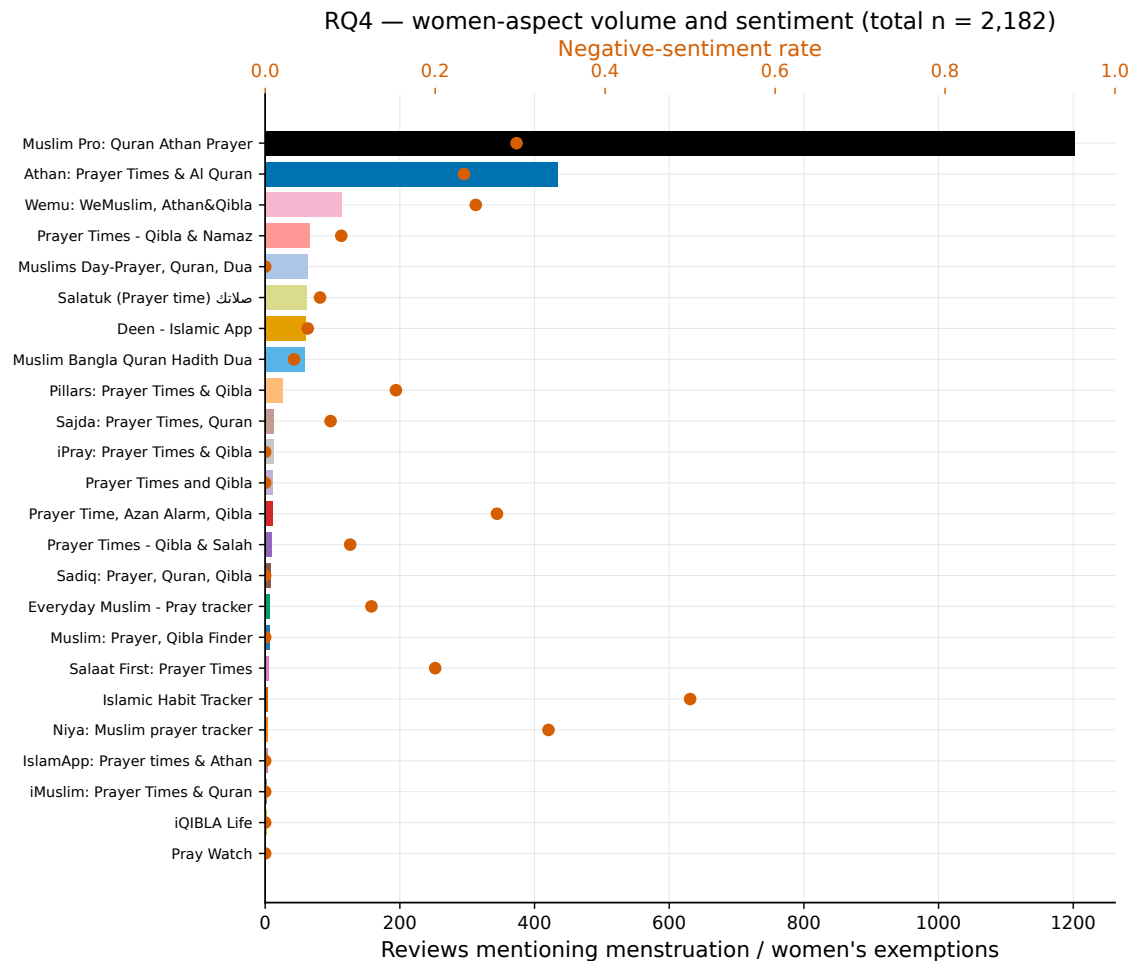


Fig. 7. Reviews mentioning menstruation or prayer exemptions, by application, with negative-sentiment rate. The three applications shipping menstrual handling are named in the text.

7.3 Having It Does Not Help

If this were purely a coverage gap, apps that ship menstrual handling should show better sentiment on the topic. They do not: 26.3% negative in the three apps that have it, against 26.0% in apps that lack it — indistinguishable. This points to execution problems, not just availability:

“The only complaint I have is: when I turn on menstrual mode it doesn’t turn off the prayers as it promises to do. Please fix it”

— Athan: Prayer Times & Al Quran, 4 stars

The RQ1 coding passes independently corroborate this: “menstrual mode unreliable” appears three times in the single-app pass, and “menstrual exemption absent” once in the cross-app pass, from a different app (§4.4).

7.4 Why This Cannot Be Tested Causally

With only three apps on the treated side, no causal claim about star rating is possible, and the numbers show why. Review-weighted, apps with the feature average 4.476 stars against 4.391 without (+0.085); averaged first by app and then across apps, the sign flips to -0.236 (4.135 against 4.371). Both are arithmetically correct and answer different questions, since one app (Athan) contributes nearly all the treated reviews. A result that reverses sign depending on how it is averaged, on three treated apps, is the clearest demonstration that $N = 3$ cannot answer this question.

A topic sub-model fit on women’s-health discourse also cannot help: six of its ten topics collapse onto a single unrelated controversy, one app’s data-sale episode, leaving no room to further resolve menstrual-handling discourse specifically (§11).

Bio-spiritual inclusion is an unmet need rather than a contested design question. `women_period` is among the four aspects under 25% ecosystem coverage and ranks second on RQ6’s unmet-need index (§9).

8 RQ5: Feature Bloat and Companion Hardware

A common claim holds that adding features creates clutter complaints, and that leaner apps are more satisfying. We tested this directly, on a hypothesis pre-specified on 2026-08-26, before running the full corpus. It fails in both directions: feature count does not predict bloat complaints, and if anything the relationship runs backward. What users call “bloat” turns out to be about the interface, not the feature count.

8.1 Feature Count Does Not Predict Bloat Complaints

The primary mixed-effects model (M4) did not converge on its `feature_count` term, so we report the cluster-robust sensitivity fit: the interaction between bloat complaints and feature count gives $\beta = -0.000487$ (95% CI $[-0.00136, 0.00038]$, $p = 0.272$), clustered on 26 apps. Adding features does not make users more likely to call an app bloated. Nor is this a popularity artefact: feature count correlates only weakly with install count across the 26 apps (Spearman $\rho = 0.244$).

8.2 If Anything, It Runs Backward

Figure 8 shows the opposite of what the bloat hypothesis predicts: the five leanest apps (6–9 features) average 3.96 stars at a 5.2% bloat-complaint rate; the five fullest (13–14 features) average 4.71 stars at 1.6% — 3.3 times lower. The pattern survives dropping the two smallest apps by review volume, and the clearest single pair is iPray (8 features, 8.0% bloat rate) against Athan (13 features, 1.5%). An exploratory app-level correlation, entered into the correction and surviving it, points the same way: feature count correlates positively with mean sentiment ($\rho = +0.471$, $p = .015$) and mean score ($\rho = +0.422$, $p = .032$).

8.3 Bloat Complaints Are Interface Complaints

So what drives them? A review complaining about bloat is 23.98 times more likely than base rate to also complain about the interface (Fisher exact OR = 29.41, $p = 7.4 \times 10^{-137}$). Even under the most conservative reading — restricting to the 40 reviews that keep the two complaints in separate sentences, rather than fusing them into one phrase — the lift is still 7.62 times (OR = 8.78, $p = 9.3 \times 10^{-24}$). Users often say it as one thing:

“Widget takes up too much space, [5×2], but displays information only in [4×1]. Remaining blocks are blank, unusable for other apps, just hogging space. Otherwise, good app, used daily. Has more than timetables which for me is just unimportant clutter.”

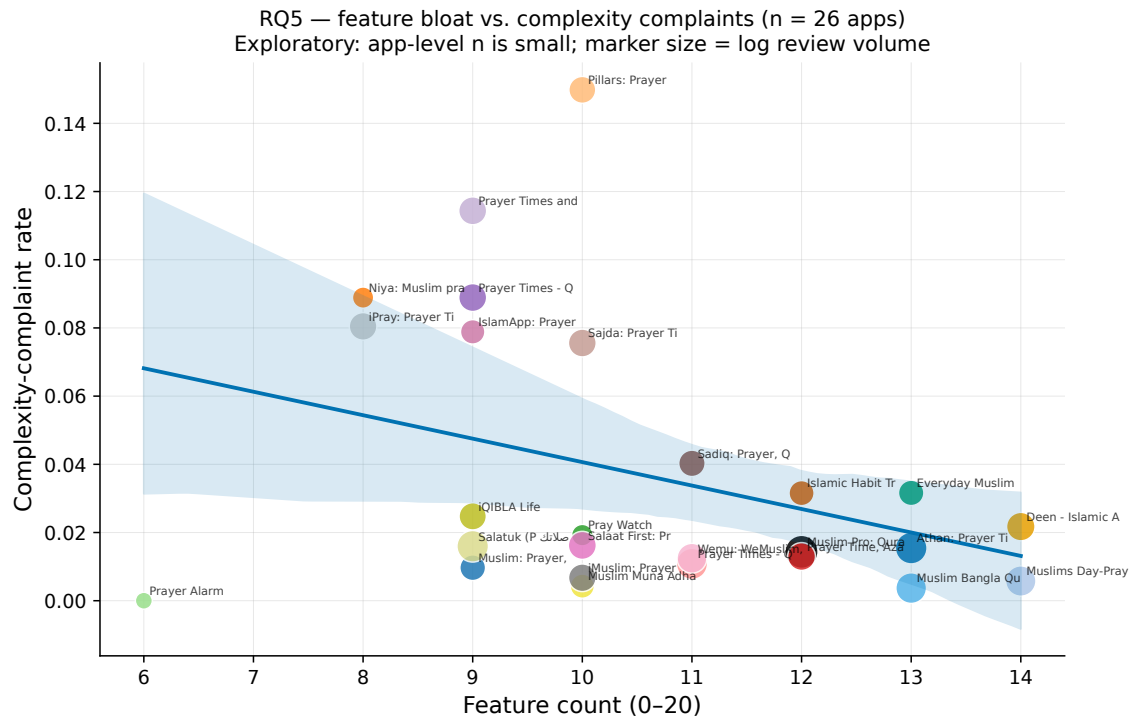


Fig. 8. Bloat-complaint rate against feature count. The leanest apps draw bloat complaints at more than three times the rate of the fullest.

— Athan: Prayer Times & Al Quran, 4 stars

A separate topic model fit on bloat-tagged reviews alone agrees, from a different angle: its top topics are dominated by users *praising* simplicity (minimal UI, no ads), with clutter, advertising, and slowness on the negative side. Users are evaluating how much the interface shows them, not how many features the app contains.

8.4 Companion Hardware: A Case of One

One app, iQIBLA Life, ships a companion hardware device. Sentiment toward hardware runs *better* inside that app (37.7% negative) than across the 23 of the 25 apps without one that mention hardware at all, where mentions are almost entirely unmet requests for a watch or ring (49.2% negative). The companion_hardware tagger runs at 40% precision, so these figures rest on tags of that quality. Shipping the hardware is associated with better sentiment, not worse — the opposite of a bloat story, and consistent with RQ3: unmet requests read more positively than usage reports of something that can fail.

The feature-bloat hypothesis fails in either direction. Feature count is uncorrelated with bloat complaints, and the leanest apps draw them at over three times the rate of the fullest. What users call bloat is a property of the interface rather than a count of capabilities.

Table 4. Features with at least one broken promise among those with $n \geq 30$ judgeable mentions and more than one promising application. `tracker_score`, with a single promising application, is discussed in the text rather than listed.

Feature	Broken / promising apps
Monetisation	5 / 16 (31%)
Reminders / adhan	5 / 20 (25%)
Mosque finder	1 / 4 (25%)
Forbidden times	1 / 4 (25%)
Widgets	2 / 13 (15%)
Prayer-time accuracy	2 / 19 (11%)
Qibla	1 / 18 (6%)

Table 5. Unmet-need ranking: demand-signal volume $\times$ applications lacking the feature.

Aspect	Score	Apps lacking
Companion hardware	6,725	25
Menstrual handling (women_period)	4,738	23
Mosque finder	3,304	14
Forbidden times	2,805	15
Nafl (supererogatory) prayer times	2,445	15

9 RQ6: Completeness and the Two Gaps

The final research question asks directly whether current apps are “enough.” The answer is no, but that single word hides two different failures that call for different remedies. A *coverage gap* is a market failure: whole categories of need are served by only a minority of apps despite measurable demand. A *delivery gap* is a quality failure, and the more damning of the two, because it falls on promises apps have already made.

9.1 The Delivery Gap

We compared each app’s store-description promises against the sentiment its reviews record for that feature. Of 272 promised (application, feature) pairs, 136 carry enough mentions to judge reliably ($n \geq 30$); of those, 18 are broken promises across 11 applications — a feature the app claims to offer that draws over 40% negative sentiment. Table 4 ranks the break rate by share of promising applications, and Figure 9 shows the full delivery gap. The eighteenth is `tracker_score`, broken in the one application that promises it; a 1-of-1 ratio carries no weight, so we leave it out of the table. The two largest broken promises by mention volume are both from Muslim Pro: its reminders/adhan feature (9,989 mentions, 45.6% negative) and its monetisation terms (9,182 mentions, 54.8% negative).

The two features broken in the most applications, five each, are also the two most basic: adhan reminders are the core function of a prayer-times app, not an advanced capability, and monetisation is simply the terms on which the app is offered. Failure here concentrates in the promise the app is built around, not in peripheral functionality.

9.2 The Coverage Gap

We ranked aspects by an unmet-need score, demand-signal volume multiplied by the number of applications lacking the feature (Table 5, Figure 10). Companion hardware tops the ranking, followed by menstrual handling, mosque finder, forbidden-times awareness, and *nafl* (supererogatory) prayer-time support. Four aspects fall below 25% ecosystem coverage altogether: calendar sync, companion hardware, qasr travel, and menstrual handling.

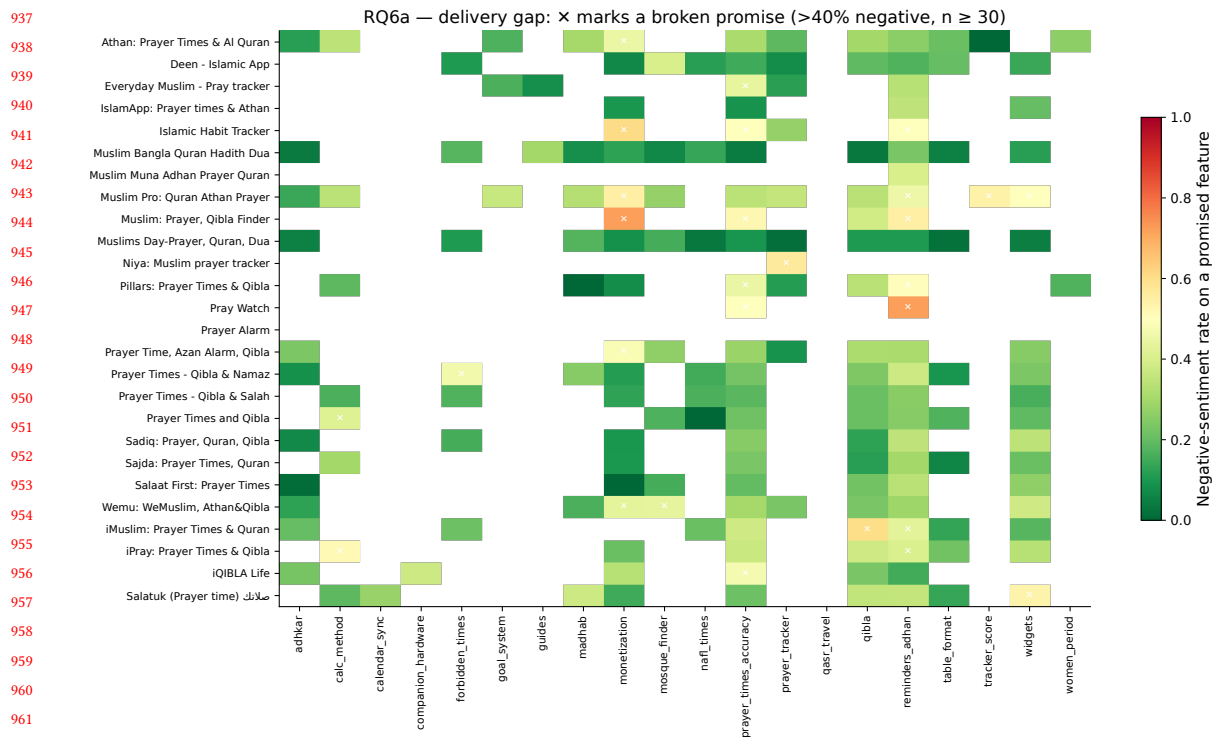


Fig. 9. The delivery gap: negative-sentiment rate per (application, feature) pair across the 136 judgeable pairs.

Five aspects clear both bars for adequate service, present in over half the ecosystem and averaging under 30% negative sentiment: madhab, monetisation, qibla, table format, and widgets. Monetisation appears on both this list and the broken-promises list — most applications handle payment acceptably, while a minority handle it badly enough to poison the feature ecosystem-wide.

9.3 The Same Gaps, Reached Independently

Each gap is also reached independently by an earlier research question, which we read as corroboration rather than repetition. Companion hardware tops the unmet-need ranking, and RQ5 finds hardware sentiment is actually *better* where it is shipped (§8) — consistent with demand rather than dissatisfaction with an existing product. Menstrual handling ranks second and is RQ4's central finding in its own right (§7). Reminders/adhan, tied with monetisation as the most-broken promise, is independently the dominant tracker-dissatisfaction theme in RQ1's cross-app coding, at 46% (§4).

9.4 Verdict

Current applications are not enough, and they fall short in two different ways. The coverage gap: entire categories of need — companion hardware, menstrual accommodation, travel concession, calendar integration — are served by under a quarter of the ecosystem despite measurable, explicitly voiced demand. The delivery gap: the two most-broken

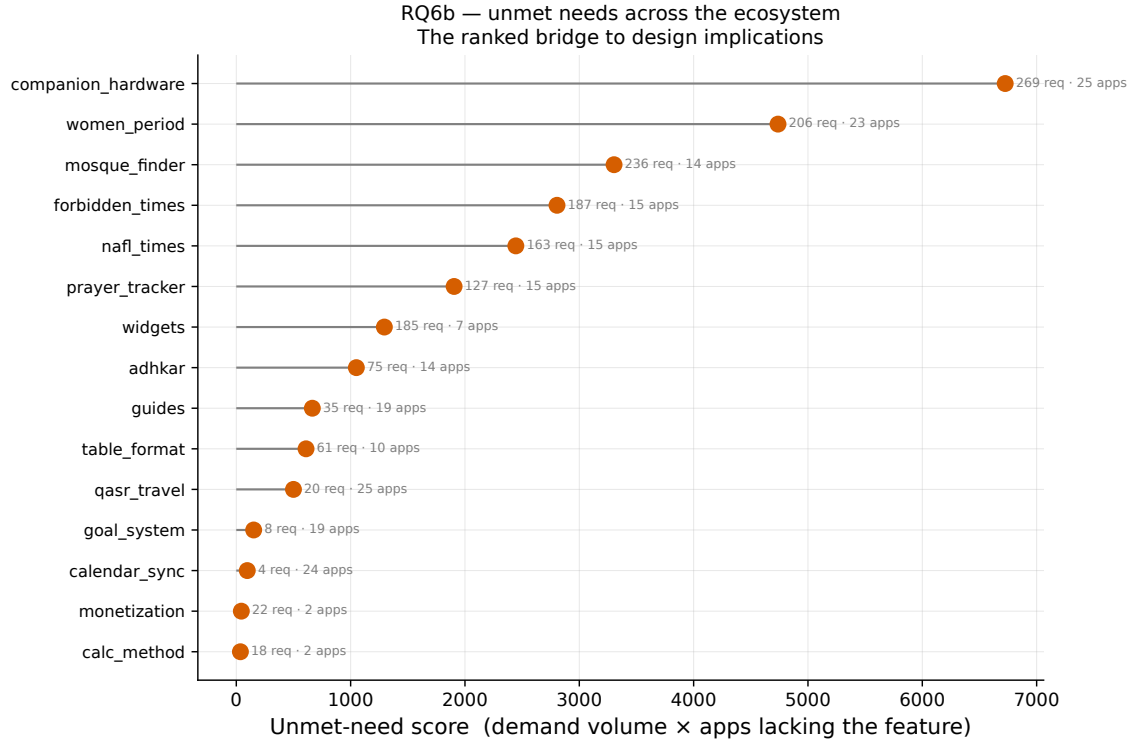


Fig. 10. Unmet-need ranking across aspects: demand-signal volume against the number of applications lacking the feature.

promises are adhan reminders and monetisation terms, the two things a prayer-times application most fundamentally offers. Building more features is not indicated. Delivering the promised ones reliably, and extending coverage to the populations currently unserved, is.

10 Discussion

One pattern recurs across all six research questions: satisfaction with a devotional app is governed by the reliability of its core functions, not by feature count or polish. Three framings borrowed from the broader HCI and self-tracking literatures mislocate where the problem sits.

10.1 Three Reframings

Gamification. The self-tracking literature predicts guilt induced by streaks and scores. That is not what we find (§4). The real harm is a record users cannot trust and a prompt that does not arrive. Where a genuine normative objection does surface, it is that the app *models worship incorrectly* — no menstrual exemption, no supererogatory (*sunnah*) prayer support. Softening a streak mechanic does not fix a tracker that does not know a user is exempt from prayer that day.

Accuracy versus aesthetics. This trade-off is mis-specified (§5). Prayer-time correctness behaves like a reliability requirement, disjoint in cost from every interface complaint and carrying the corpus's second-highest penalty, while

madhab configuration behaves like an ordinary preference, cheap to get wrong. Treating “accuracy” as a single design dimension hides exactly the distinction that matters for prioritisation.

Feature bloat. What users call bloat is a property of what the interface *surfaces* in the moment, not of how many capabilities the application *contains* (§8). The feature-count framing cannot express that distinction, which is why it fails in both directions at once.

10.2 The Two Gaps

RQ6 (§9) names two failures that the other five research questions independently arrive at from different directions. Coverage is a market failure: whole categories of need are served by under a quarter of the ecosystem despite demand that is explicit and easy to find in review text. Delivery is a quality failure, and the more consequential, because it falls on promises applications have already made. The two most-broken promises — adhan reminders and monetisation terms — are the core function of a prayer-times app and the terms it is offered on.

10.3 Methods Lessons

Two of this paper’s four contributions are methodological.

Star ratings are an unsafe proxy for defect severity. 41.6% of reviews carrying a negative accuracy complaint award four or five stars (§5). Text, not stars, is where this class of complaint is visible.

Single-app qualitative sampling can manufacture genre-level themes. Themes that looked like general properties of gamified prayer tracking accounted for 51% of a single-application coding pass and only 4% of an independently drawn, stratified cross-application pass on the same phenomenon (§4.4).

Devotional register breaks off-the-shelf NLP. A feature-vocabulary match scored 0% precision because reviewers routinely close reviews with formulae such as “May Allah reward you,” which generic keywords mistake for feedback about a tracking feature (§3.2).

10.4 Threads That Recur

Advertising and monetisation run together through the results without being the same thing: intrusive advertising is the costliest complaint in the corpus by coefficient (§5) and a dominant theme in bloat discourse (§8), while monetisation, tracked separately, is an RQ1 coding theme and one of RQ6’s two most-broken promises. Adhan reminders are the most-cited reason users give for choosing an app (§6), one of the two most-broken promises (§9), and the dominant tracker-dissatisfaction theme at 46% (§4). Beneath both sits one want: an accurate, owned, accommodating record of one’s own worship — the record, not the streak.

11 Limitations

- **No users were studied.** Every finding comes from public app-store reviews, a self-selected sample skewed toward strong positive and strong negative experiences. Claims here are claims about what reviewers chose to write down, not direct claims about religious practice.
- **The corpus is English/US-locale and purposively sampled.** It under-represents the regions where most of the world’s Muslims live (§3.1). RQ2’s finding that madhab complaints are mild and cheap should not be read as a global claim.

- **Aspect-tagging precision is uneven.** Seven of the 25 aspects with surviving gold rows fall below 50%, two at 0% (§3.6). Every count for these seven is reported elsewhere as a precision-adjusted estimate with its interval, never as a raw tagger count.
- **Eleven model terms cannot support inference.** Eight are non-converged rather than precisely estimated nulls, kept in the Benjamini–Hochberg denominator of 29 for transparency; three more return no error estimate at all and sit outside it (§3.5). Every null reported here is cited from a fit that converged.
- **The topic model covers a minority of the corpus.** 22.7% of modelled documents fall in an outlier bucket and 61% in one undifferentiated praise cluster; the 39 labelled topics cover about 16%.
- **The RQ4 sub-model collapsed around one application’s privacy episode.** Six of ten topics concern one application’s user data being sold to US military-linked buyers, a 2020 controversy specific to Muslim Pro, leaving no room to further resolve menstrual-handling discourse (§7).
- **Illustrative quotations skew toward one application.** Most of the reviews quoted in this paper come from Athan, the corpus’s largest tracker-bearing application, because quotation followed the volume of available text rather than a sampling rule — the same hazard we identify as a methodological contribution (§4.4).
- **The affect lexicon is unvalidated.** The guilt and motivation keyword lists behind RQ1’s lexical comparison have no gold sample (§3.2). RQ1’s claim about guilt rests on the hand-coded passes, which point the same way.
- **Two qualitative coding passes, never pooled,** coding 80 reviews between them — supporting evidence, not a primary contribution.
- **Three small-*N* case studies are descriptive only.** Inclusion features ship in 3 of 26 applications; companion hardware and qasr travel concession each ship in exactly 1. None supports a between-app statistical estimate.
- **Promise-source agreement is weak.** Store descriptions and the hand-built feature matrix agree at a mean Cohen’s κ of 0.272, against 73.1% mean raw agreement (§3.3). Any claim depending on what an application promises should be read with this ceiling in mind.
- **The demand-precision sample predates the final extraction.** The 200-row gold sample behind the 91.5% figure was labelled against an earlier run (§3.3).
- **Topic modelling covers English-identified text only.** Findings do not characterise discourse in the languages this filter excluded.
- **Reviews are cross-sectional, not longitudinal.** Nothing links a user’s reviews across time, so we can speak to aggregate patterns but not to how an individual’s relationship with an application changes over their tenure.

12 Design Implications

We do not recommend adding features generically: feature count is not the lever satisfaction runs on. Each recommendation ties to a specific reported result, and every coverage recommendation pairs with its delivery point, since the two gaps need different remedies.

12.1 Deliver What Is Already Promised, Before Adding Anything

The two most-broken promises are also the two most basic: adhan reminders (broken in 5 of 20 promising applications) and monetisation terms (5 of 16) (§9). Prompting failure is independently the dominant tracker complaint, at 46% of the cross-app coding pass (§4). An application that cannot reliably fire a notification cannot support a prayer-tracking practice built on top of it, whatever else it ships.

12.2 Treat Prayer-Time Correctness as a Reliability Requirement

Prayer-time accuracy costs −1.112 stars, the second-costliest of the complaint aspects we model, and 41.6% of accuracy complaints arrive inside four- or five-star reviews (§5). A team prioritising defects from its own rating distribution will systematically under-prioritise this class of failure. We recommend instrumenting for accuracy defects directly, comparing computed times against an authoritative reference, rather than inferring severity from ratings that conceal them.

12.3 Accommodate Practice, Rather Than Softening Mechanics

Where tracker dissatisfaction carries a normative complaint, it is that the app mis-models worship, not that its mechanics apply too much pressure (§4). Menstrual handling makes this concrete: it ranks second on the unmet-need index, is absent from 23 of 26 applications, and in the three that do ship it, sentiment is no better than in applications without it (§7). This is simultaneously a coverage problem and a quality problem; closing the coverage gap without fixing reliability would not resolve it.

12.4 Surface Less; Do Not Necessarily Contain Less

The lever available to a design team is what the interface surfaces in a given moment, not how many capabilities the application contains (§8). A full-featured application with a disciplined interface is not the problem the bloat framing warns against; a lean application that surfaces its few features poorly can still earn the complaint.

12.5 The Unserved Categories

Four categories fall below a quarter of ecosystem coverage against demonstrated demand (§9). **Companion hardware** tops the ranking, absent from 25 of 26 applications, and sentiment toward hardware is *better* inside the one application shipping a device (§8), so this is demand rather than dissatisfaction with an existing product. **Menstrual accommodation** ranks second (§12.3). **Travel concession** (*qasr*) has essentially no market presence (§6), and **calendar integration** rounds out the four. None of these calls for adding features indiscriminately; each is a demand-evidenced gap in an ecosystem whose more pressing problem is delivering what it has already promised.

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